

AI4I 2020 — Predictive Maintenance

Exploratory Data Analysis Report

Prepared by: **Win** | Project #1 of 3 — Industrial Data Analyst Portfolio | May 2026

1. Executive Summary

This report explores the AI4I 2020 dataset — 10,000 records of synthetic but realistic machine sensor readings designed to mirror real predictive-maintenance scenarios. The goal: understand how, when, and why machines fail, and turn that understanding into a foundation for a predictive model and a Looker Studio dashboard.

Top-line numbers

Records analyzed

10,000

0 missing, 0 duplicates

Failure rate

3.39%

339 of 10,000 machines

Most common cause

HDF

Heat Dissipation — 33.9%

Highest-risk variant

L (Low)

3.92% fail rate

Bottom line

Failures are rare (3.4%) but predictable. Heat dissipation issues drive a third of all failures, and lower-grade machines fail almost twice as often as high-grade ones.

2. Key Findings

Finding 1 — Failures are rare but real

Out of 10,000 machine cycles, 339 ended in failure. That is a 3.39% failure rate.

Overall Machine Failure Rate

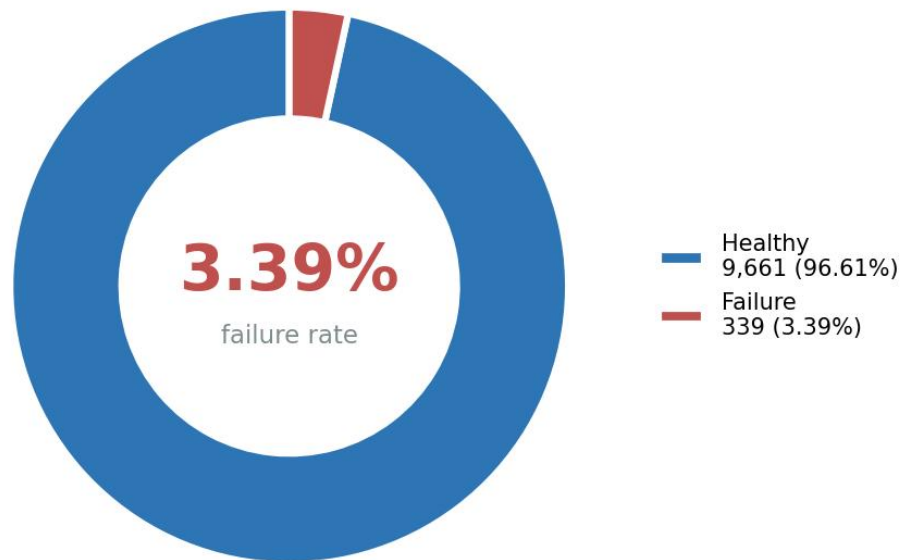


Figure 1 — Overall machine failure rate (339 / 10,000)

Finding 2 — Heat dissipation is the single biggest cause

When a machine does fail, the most likely reason is HDF — 115 of the 339 failures.

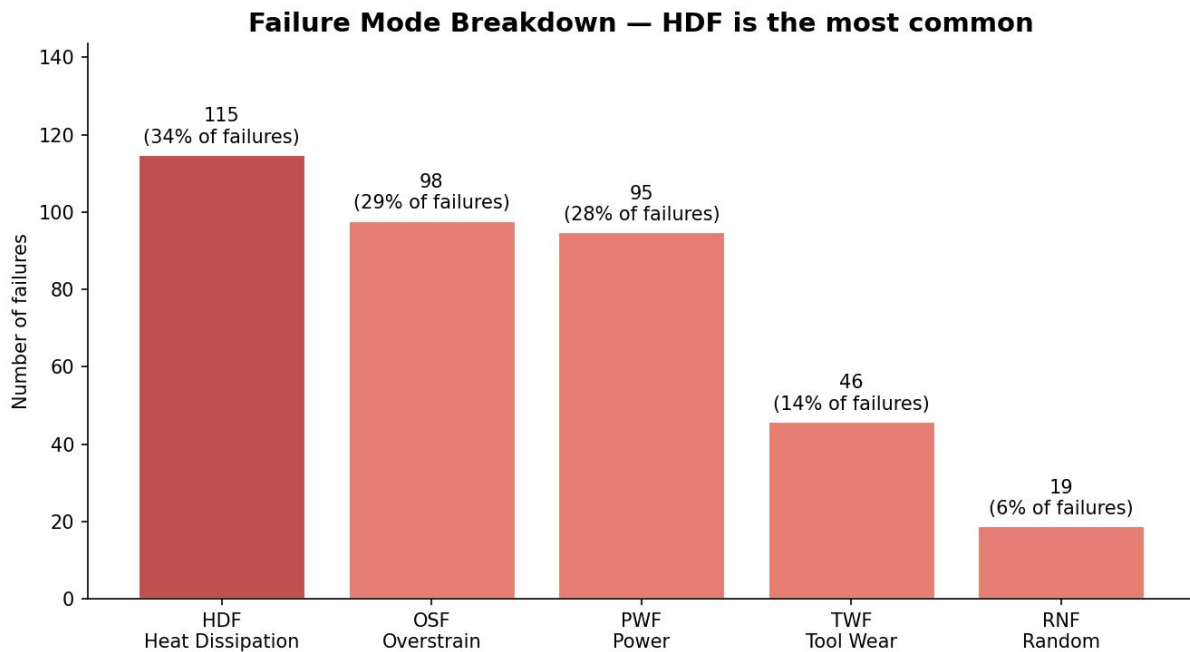


Figure 2 — Failure breakdown by cause

Finding 3 — Lower-quality machines fail nearly 2× more

L (Low) variant fails at 3.92% — almost double the H (High) variant's 2.09%.

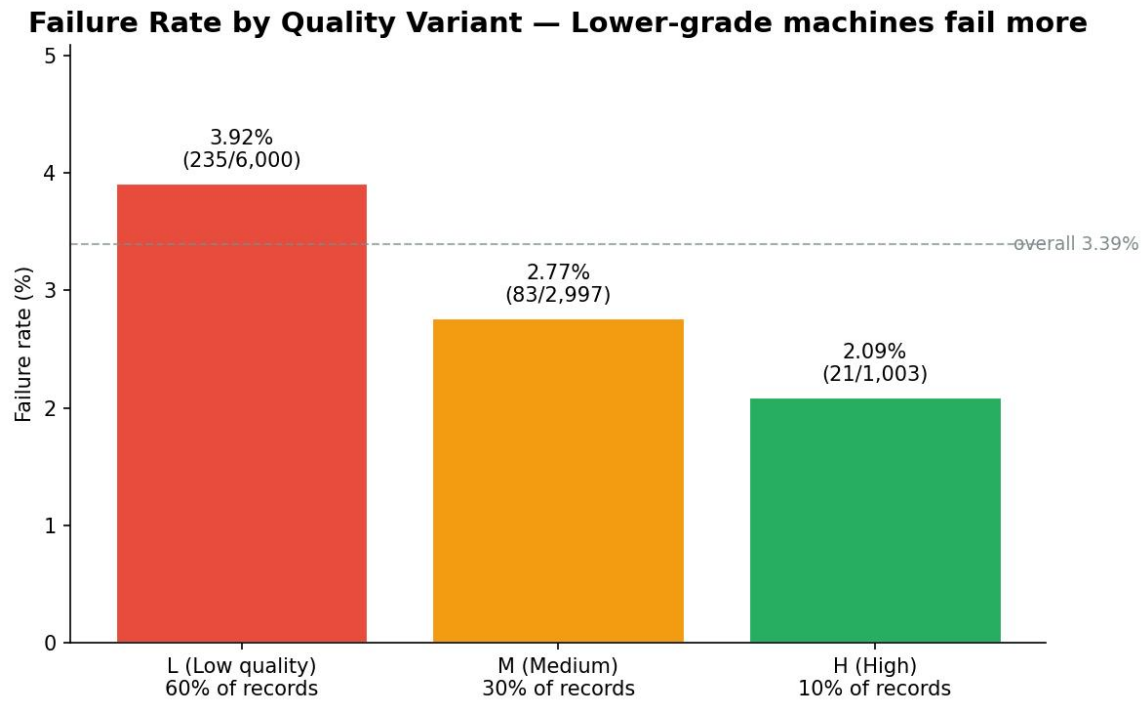


Figure 3 — Failure rate by quality variant

Finding 4 — Failures cluster in a clear operating window

Failures concentrate where torque is high AND rotational speed is low — the danger zone.

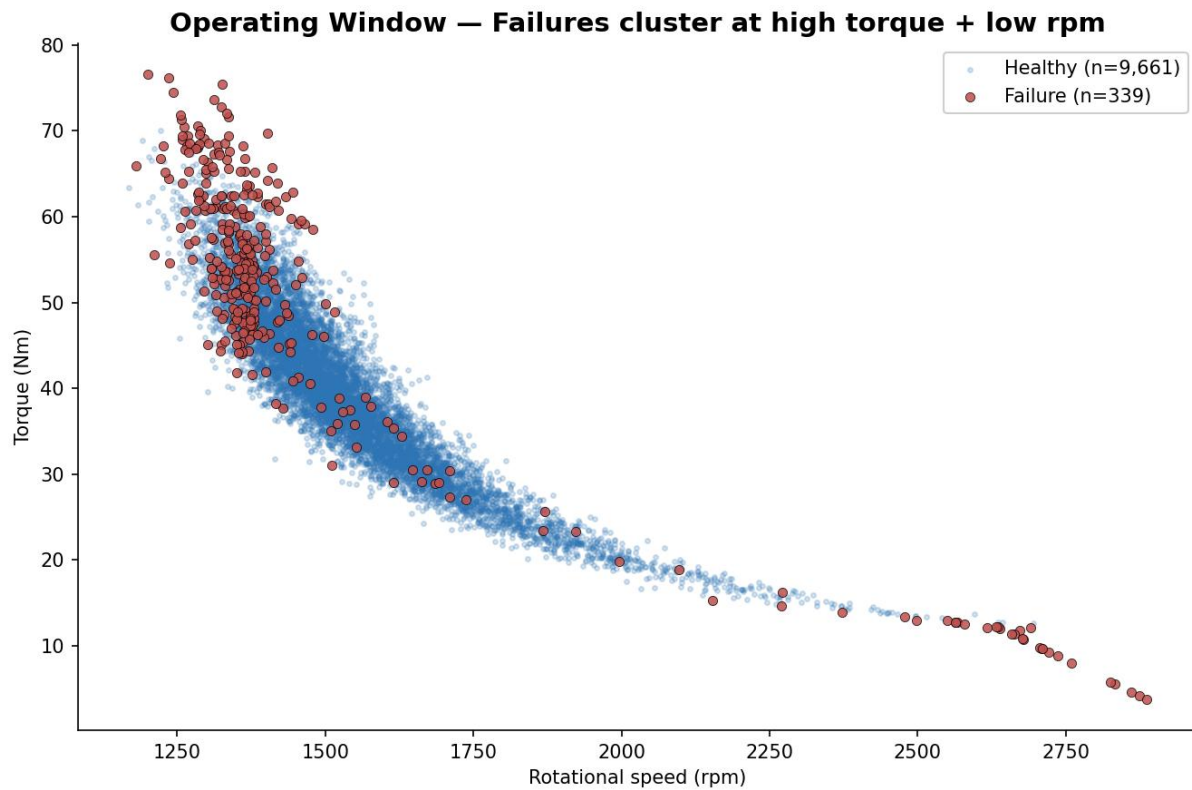


Figure 4 — Operating window

Operational rule of thumb

If torque > 60 Nm AND rotational speed < 1,400 rpm, the machine is in the high-risk zone — 40% failure rate vs 3.4% baseline.

Finding 5 — No single sensor predicts failure alone

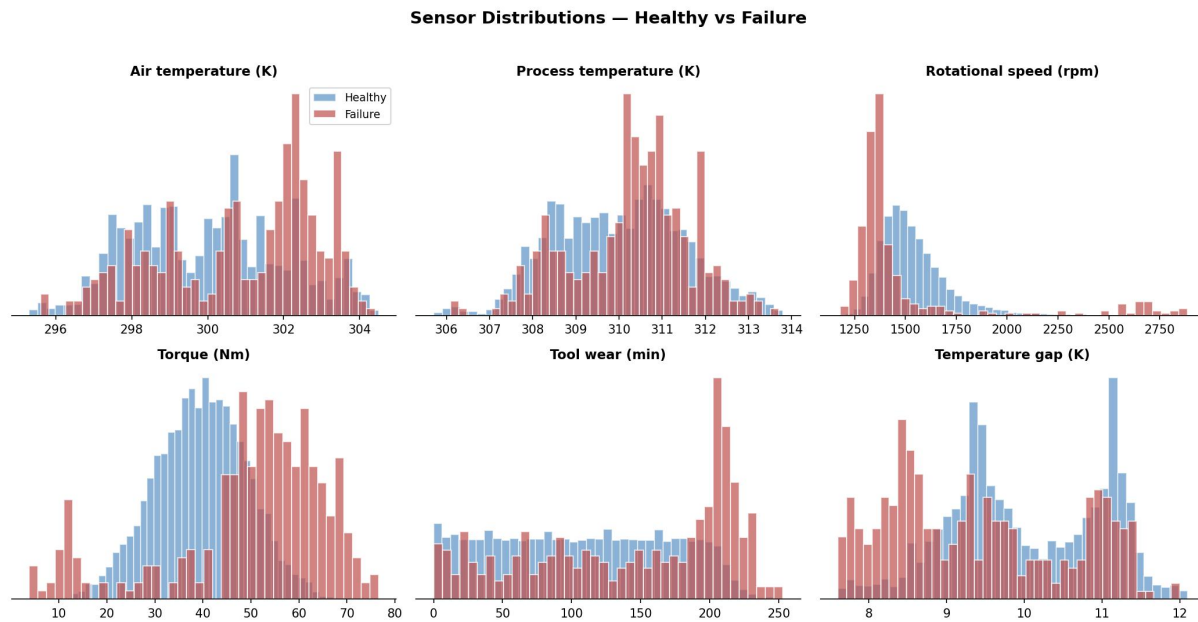


Figure 5 — Sensor distributions: healthy vs failure

3. Business Impact

A predictive model targeting HDF + OSF + PWF could catch ~85% of all failures. If a Thai factory loses THB 50,000 per hour of unplanned downtime and currently averages 8 failures per day at 30 minutes each, that is THB 200,000 per day in losses. A model preventing half of those failures saves ~THB 3M per month per line.

4. Next Steps

1. Build Looker Studio dashboard from data/processed/ai4i_clean.csv
2. Project #2 — Predictive ML model (Random Forest / XGBoost)
3. Apply to NEXPIE, Siemens Thailand, PTT/IRPC